



PES University, Bengaluru
(Established under Karnataka Act 16 of 2013)

END SEMESTER ASSESSMENT (ESA) - DEC 2025

UE22CS341B - Compiler Design

Total Marks : 100.0

1.a. With a diagram depict the different phases of a Compiler with appropriate Input and Output specified for each phase. (10.0 Marks)

1.b. With an example describe the various error-recovery techniques used in syntax analysis. (5.0 Marks)

1.c. Given a grammar $S \rightarrow S S \mid a$
i) Is it left recursive?
ii) Can it be converted into LL(1) grammar?
iii) Justify your answer (5.0 Marks)

1.d. Construct first and follow set for the given grammar.

$S \rightarrow a \mid (L)$

$L \rightarrow L,S \mid S$

(5.0 Marks)

2.a. Build LR(0) for the following grammar and identify whether the grammar is in LR(0)

$X \rightarrow Yz \mid a$

$Y \rightarrow bZ \mid \lambda$

$Z \rightarrow \lambda$

(8.0 Marks)

2.b. Write an SDD to count the number of bits in a binary number using suitable grammar and semantic rules. (5.0 Marks)

2.c. Shift-Reduce Parsing: Given the following ACTION/GOTO table, show the parse if the current stack contents are 0 b 4 B 1 (i.e., current state is 1) and the remaining input is "d\$".

State	ACTION			GOTO	
	d	e	\$	S	B
0	Shift 2	Reduce $S \rightarrow Bb$	Reduce $S \rightarrow bB$	2	1
1	Shift 3	Reduce $B \rightarrow d$	Accept	3	4
2	Shift 1	Shift 4	Accept	2	4
3	Reduce $S \rightarrow Bd$	Reduce $B \rightarrow d$	Reduce $S \rightarrow bBd$	1	2
4	Reduce $B \rightarrow e$	Shift 3	Reduce $S \rightarrow bSe$	4	3

(5.0 Marks)

2.d. Provide a production with the shortest RHS and explain how that will introduce s/r conflict in SLR parser for the following grammar

$S \rightarrow Ab$

$A \rightarrow cbAd$

$A \rightarrow cAd$

$A \rightarrow \lambda$

(7.0 Marks)

3.a. Complete the semantic rule for the following L-attributed SDD. Evaluate the SDD for the input 3+4*5

$E \rightarrow T E'$

$E' \rightarrow + T E'_1$

$E' \rightarrow \lambda$

$T \rightarrow F T'$

$T' \rightarrow * F T'_1$

$T' \rightarrow \lambda$

$F \rightarrow \text{num}$

(8.0 Marks)

3.b. Write the semantic rules and the translation scheme for the following grammar representing an if statement:

$S \rightarrow \text{if}(C)S_1$ (6.0 Marks)

3.c. Generate Three-address code for the following program, express the same in Quadruple and Triple representation

```
int sumArray(int A[10], int n) {  
    int i, sum = 0;  
    for (i = 0; i < n; i++) {  
        sum = sum + A[i];  
    }  
    return sum;  
}
```

(6.0 Marks)

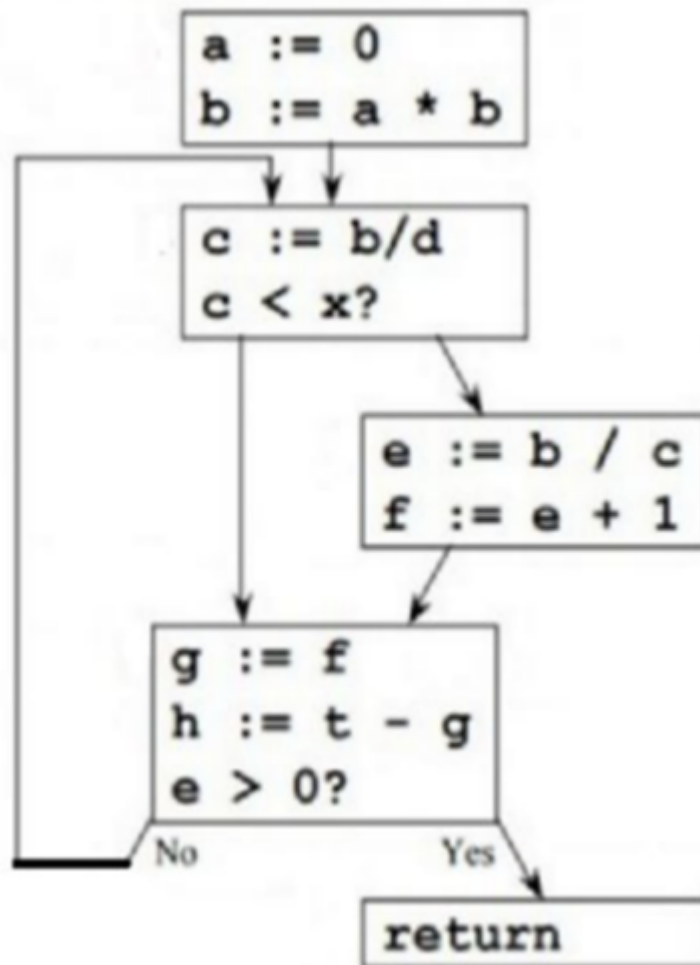
3.d. Convert the given intermediate code to SSA.

```
i=1  
j=1  
k=0  
while k<100  
    if j<20  
        j=i  
        k=k+1  
    else  
        j=k  
        k=k+2  
return j
```

(5.0 Marks)

4.a.

Calculate liveness information for the given control flow graph



(5.0 Marks)

4.b. Explain the terms with an example

- a) Strength Reduction
- b) Packing Temporaries
- c) Loop invariant Detection
- d) Loop Invariant Code motion

(6.0 Marks)

4.c. Discuss the issues in the design of the code generator. (7.0 Marks)

4.d. Construct target code for the following procedures assuming stack allocation:

main()

$x = y + 2$

 call foo

foo()

$z = x * x$

 return

Assume the following:

1. The stack area starts at location 800
2. The code area for main() starts at address 100
3. The code area for foo() starts at address 400
4. The activation record size of main() is 60 bytes
5. The activation record size of foo() is 30 bytes

(7.0 Marks)